

Challenge :- The main challenge was handling JPEG compression artifacts. MS Paint images are pure black and white, but saving as .jpg introduces slight colour variations. A pure black pixel can become "almost black" with mixed RGB values. A simple exact colour match would miss these pixels and return the wrong count.

To solve this, instead of checking if a pixel is exactly black, I calculate its brightness using a standard luminance formula ($Y = 0.299R + 0.587G + 0.114B$) and treat anything below a threshold as black. This approach is more robust and handles any compression noise reliably.

I also considered other approaches such as edge detection algorithms or histogram-based methods, but chose column scanning with a brightness threshold because it is simple, easy to understand, and sufficient for clean MS Paint images.